PRODUCT REQUIREMENT DOCUMENT (PRD)

Finite Automata

UMHackathon 2026

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Project Name: AI Fridge – Smart Grocery & Food Waste Reduction System

Version: 2.0

1. Project Overview

- **Problem Statement:** Households waste approximately **30–40% of purchased food** due to poor inventory tracking, forgotten groceries, and lack of visibility into expiration dates. Existing solutions fail because they rely heavily on **manual input**, leading to user drop-off within days. Additionally, grocery data (receipts), kitchen inventory, and meal planning tools exist in silos, creating a fragmented and inefficient experience.

- **Target Domain:** AI-powered household automation, smart kitchen management, and sustainability-focused consumer applications.

- **Proposed Solution Summary:** Fridge AI is an **agentic AI system** that automates the entire grocery lifecycle—from purchase to consumption. By using **Z AI + OCR + NLP + multi-agent workflows**, the system extracts items from receipts, tracks inventory states, predicts expiration, and recommends recipes using available ingredients. This eliminates manual effort and creates a **closed-loop "purchase-to-pantry-to-plate" ecosystem**.

2. Background & Business Objective

- **Background of the Problem:** Most food waste happens because people simply forget what they already own. Existing inventory apps require manual entry, which users abandon after days. Receipts, fridge contents, and meal plans are not connected, forcing users to juggle multiple tools or rely on memory. This leads to duplicate purchases, spoiled food, and wasted money.

- **Importance of Solving This Issue:** Reducing household food waste by even 50% would significantly lower environmental impact (less landfill methane) and save families hundreds of dollars annually. An automated, low-friction solution can change daily habits without requiring user discipline.

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- **Strategic Fit / Impact:** This aligns with AI-driven automation for everyday life. By combining OCR, multi-agent state management, and personalized recommendations, the app demonstrates practical agentic workflows that scale across logistics, retail, and smart home ecosystems.

3. Product Purpose

3.1. Main Goal of the System: To automatically track household groceries from purchase to consumption, reducing food waste through intelligent inventory management and recipe suggestions.

3.2. Intended Users (Target Audience):

- Busy professionals
- Families & households
- Students living independently
- Health-conscious users
- Sustainability-focused consumers

4. System Functionalities

4.1. Description: The system ingests unstructured inputs (receipt images, voice notes, text descriptions), extracts grocery items via OCR and NLP, manages their lifecycle states, and proactively suggests recipes or shopping actions via Z AI.

4.2. Key Functionalities:

- **Unstructured Input Ingestion:** Accepts crumpled/faded receipt images, voice memos ("bought apples and milk"), and typed lists.

- **Multi-Agent Coordination:** Dedicated agents for OCR, inventory state management, recipe matching, and push notifications.

- **State Decision-Making:** Tracks each item through purchased → stored → expiring_soon → expired → consumed, with automated transitions.

- **Structured Actionable Output**: JSON inventory lists, recipe cards (with matched vs. missing ingredients), shopping list generation.

4.3. AI Model & Prompt Design

4.3.1. Model Selection: Uses Z AI GLM for OCR + reasoning due to its long context window (handles multi-page receipts), fast inference, and native multimodal support. Combined with a fine-tuned BERT-based classifier for expiration estimation.

4.3.2. **Prompting Strategy:** Multi-step agentic prompting. The OCR Agent uses few-shot examples of messy receipts. The Inventory Agent uses chain-of-thought to infer item ambiguity ("milk" → defaults to 2%, user can override). Recipe Agent uses zero-shot for creative matches.

4.3.3. **Context & Input Handling**: Max receipt image size: 10MB. For oversized or blurry inputs, the system returns a ocr_low_confidence flag and asks user to retake photo or type manually. No silent truncation. Long receipts are chunked by line items.

4.3.4. **Fallback & Failure Behavior:** If OCR returns gibberish or hallucinated items, the system shows a graceful error: *"Could not read this receipt clearly. Please type the items or retake photo."* A retry button triggers a different angle suggestion. No human escalation in MVP; user can manually add items.

5. User Stories & Use Cases

- **User Stories:**
  - *"As a busy parent, I want to scan a grocery receipt so that my fridge inventory updates automatically without me typing anything."*
  - *"As someone who hates wasting food, I want to receive a notification when milk expires tomorrow so I can use it in a recipe."*

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- *"As a home cook, I want to see recipes that use only what I already have, so I don't need to buy extra ingredients."*

- **Use Cases (Main Interactions):**

  - **Receipt Scan → Inventory Update Flow:**
    Scan Receipt →Extract Items From Receipt → View and Manage Inventory → System updates state to stored → User receives confirmation → (Alternative) Add Manually if no receipt.
  - **Expiration and Alert Flow**
    System Clock triggers daily check → Auto Expire (moves past-date items to waste) → Expiration Alert for items expiring soon
  - **Expiring Soon → Recipe Suggestion Flow:**
    Expiration Alert → Get recipe suggestion → Generate Recipe Suggestion → Notification Agent sends recipe card → User clicks to view steps + missing ingredients.
  - **Shopping List Generation**
    Generate Shopping List (manual or triggered by low inventory / recipe missing ingredients) → Inventory updates after restock.

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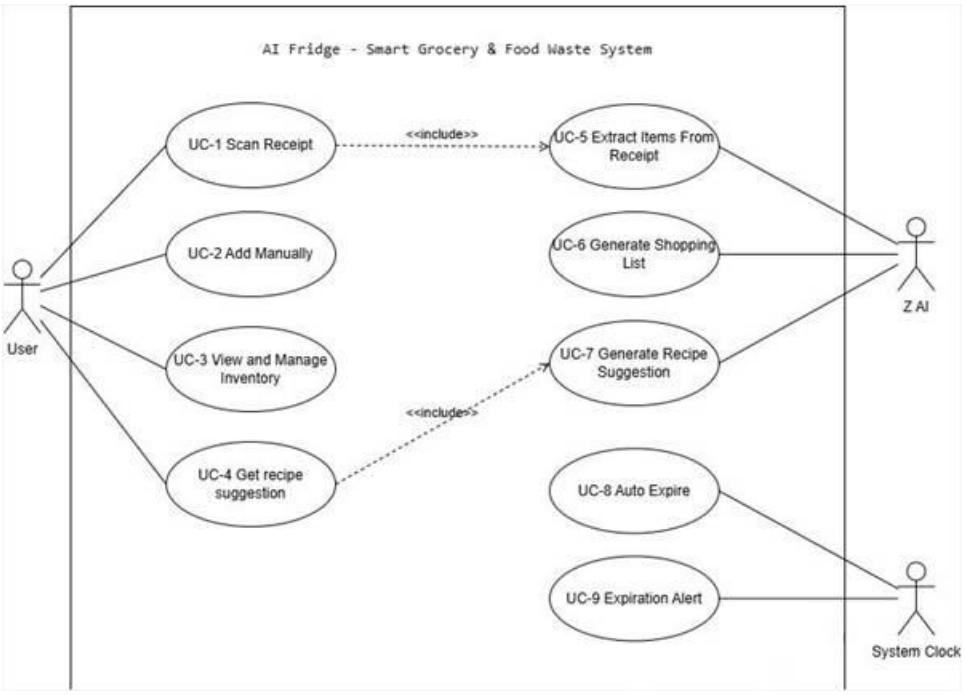
- **Use Case Diagram:**



6. Features Included (Scope Definition)

- Automatic receipt OCR + NLP item extraction (supports blurry, crumpled, or faded images)
- Inventory state machine (purchased → stored → expiring_soon → expired → consumed)
- Expiration notifications (push or in-app)
- Recipe suggestions based on available inventory + missing ingredients shopping list
- Manual override for ambiguous items ("milk" = 2% default)

7. Features Not Included (Scope Control)

- No multi-user household sharing (single user only for MVP)
- No integration with smart fridges or IoT sensors
- No backend database beyond local storage (MVP uses IndexedDB or Firebase basic)
- No automated grocery ordering / delivery integration

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8. Assumptions & Constraints

· **LLM Cost Constraint:** Estimated token cost per user session: ~50,000,000 tokens (receipt OCR + recipe suggestion). Design decision to keep costs manageable: Cache frequently purchased items (e.g., milk, eggs) so their expiration defaults and recipe mappings are stored locally, avoiding repeated LLM calls.

· **Technical Constraints:** Recipe matching is rule-based + embedding similarity fallback; no real-time grocery price API. Output format is JSON and simple recipe cards (no live cooking integration).

· **Performance Constraints:** OCR on high-res receipts may take 2-4 seconds. Expiration check runs daily (not real-time). Recipe suggestions limited to 5 per request to avoid token blow.

· **User Input:** System cannot generate inventory without a receipt or user entry. User must approve ambiguous items (e.g., "milk type?"). Not fully autonomous — requires occasional manual confirmation.

9. Risks & Questions Throughout Development

- **OCR Ambiguity:** How do we handle items like "fresh produce" (no specific name) or "bakery item" (generic)? → *Default to "unknown produce" and ask user to clarify once; after 3 skips, log as generic.*
- **Expiration Estimation Accuracy:** How do we avoid users trusting wrong dates? Always show "estimated" badge and allow manual edit. Use conservative defaults (e.g., milk = 7 days from purchase).
- **Recipe Quality:** What if the AI suggests weird combinations (e.g., "eggs + orange juice omelette")? → *Use a whitelist of tested recipe templates for MVP; generative recipe is a stretch goal.*
- **User Abandonment Again:** Even with automation, how do we keep users returning? *Weekly waste report: "You saved $X this week by using expiring items." Gamify with streak tracking.*
- **Frame Stability:** After OCR, if an item list is malformed (e.g., quantity missing), does the user need to fix manually? → *Yes. System flags* needs_review *and highlights problematic lines.*

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